

Q1 JEE Main 2020 - 7 Jan (Evening)

Let $X = \{n \in \mathbb{N} : 1 \leq n \leq 50\}$. If $A = \{x \in X : n \text{ is a multiple of } 2\}$; $B = \{n \in X : n \text{ is a multiple of } 7\}$, then the number of elements in the smallest subset of X containing both A and B is

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Q2 JEE Main 2020 - 8 Jan (Evening)

Let S be the set of all real roots of the equation, $3^x (3^x - 1) + 2|3^x - 1| + |3^x - 2|$. Then S :

- (A) singleton set
- (B) two elements
- (C) at least four elements
- (D) infinite elements

Q3 JEE Main 2020 - 9 Jan (Evening)

If $A = \{x \in \mathbb{R} : |x| < 2\}$ and $B = \{x \in \mathbb{R} : |x - 2| \geq 3\}$; then:

- (A) $A - B = [-1, 2)$
- (B) $A \cup B = \mathbb{R} - (2, 5)$
- (C) $B - A = \mathbb{R} - (-2, 5)$
- (D) $A \cap B = (-2, -1)$

Q4 JEE Main 2020 - 2 Sep (Morning)

If $R = \{(x, y); x, y \in \mathbb{Z}, x^2 + 3y^2 \leq 8\}$ is a relation on the set of integers $\mathbb{Z}$, then the domain of R^{-1} is:

- (A) $\{0, 1\}$
- (B) $\{-2, -1, 1, 2\}$
- (C) $\{-1, 0, 1\}$
- (D) $\{-2, -1, 0, 1, 2\}$

Q5 JEE Main 2020 - 3 Sep (Evening)

Let R_1 and R_2 be two relation defined as follows:

$$R_1 = \{(a, b) \in \mathbb{R}^2 : a^2 + b^2 \in \mathbb{Q}\} \text{ and}$$

$$R_2 = \{(a, b) \in \mathbb{R}^2 : a^2 + b^2 \in \mathbb{Q}\}, \text{ where } \mathbb{Q} \text{ is the set of all rational numbers. Then:}$$

- (A) Neither R_1 nor R_2 is transitive.
- (B) R_2 is transitive but R_1 is not transitive.
- (C) R_1 and R_2 are both transitive.

(D) R_1 is transitive but R_2 is not transitive.

Q6 JEE Main 2020 - 4 Sep (Morning)

A. Survey shows that 63% of the people in a city read newspaper A whereas 76% read newspaper B. If $x\%$ of the people read both the newspapers, then a possible value of x can be :

- (A) 37
- (B) 55
- (C) 29
- (D) 65

Q7 JEE Main 2020 - 4 Sep (Evening)

Let $\bigcup_{i=1}^{50} X_i = \bigcup_{i=1}^n Y_i = T$, where each X_i contains 10 elements and each Y_i contains 5 elements. If each element of the set T is an element of exactly 20 of sets X_i s and exactly 6 of sets Y_i s then n is equal to

- (A) 50
- (B) 15
- (C) 30
- (D) 45

Q8 JEE Main 2020 - 5 Sep (Morning)

A survey shows that 73% of the persons working in an office like coffee, whereas 65% like tea. If x denotes the percentage of them, who like both coffee and tea, then x cannot be :

- (A) 63
- (B) 36
- (C) 38
- (D) 54

Q9 JEE Main 2020 - 5 Sep (Evening)

Let $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$. Then the number of elements in the set $C = \{f : A \rightarrow B \mid 2 \in f(A) \text{ and } f \text{ is not one-one}\}$ is

Q10 JEE Main 2020 - 6 Sep (Morning)

Set A has m elements and Set B has n elements. If the total number of subsets of A is 112 more than the total number of subsets of B , then the value of $m \cdot n$ is

Answer Key

Q1 (29)

Q2 (A)

Q3 (C)

Q4 (C)

Q5 (A)

Q6 (B)

Q7 (C)

Q8 (B)

Q9 (19)

Q10 (28)

Q1 mathongo // mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

$$\begin{aligned} n(A \cup B) &= n(A) + n(B) - n(A \cap B) \\ &= 25 + 7 - 3 \\ &= 29 \end{aligned}$$

Q2 mathongo // mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

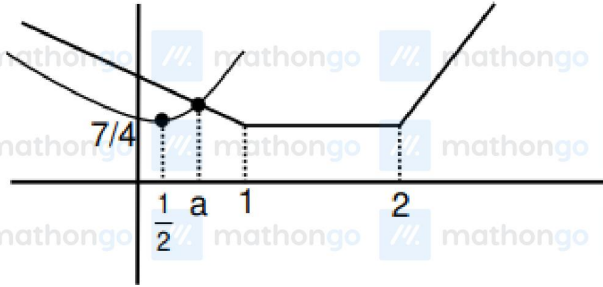
$$\begin{aligned} \text{Let } 3^x &= t \\ t(t-1) + 2 &= |t-1| + |t-2| \\ t^2 - t + 2 &= |t-1| + |t-2| \end{aligned}$$

are positive solution

$$t = a$$

$$3^x = a$$

$x = \log_3 a$ so singleton set



Q3 mathongo // mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

$$A = \{x : x \in (-2, 2)\}$$

$$B = \{x : x \in (-\infty, -1] \cup [5, \infty)\}$$

$$A \cap B = \{x : x \in (-2, -1]\}$$

$$A \cup B = \{x : x \in (-\infty, 2] \cup [5, \infty)\}$$

$$A - B = \{x : x \in (-1, 2)\}$$

$$B - A = \{x : x \in (-\infty, -2] \cup [5, \infty)\}$$

Q4 mathongo // mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

$$\text{Given } R = \{(x, y) : x, y \in \mathbb{Z}, x^2 + 3y^2 \leq 8\}$$

$$\text{So } R = \{(1, 1), (2, 1), (1, -1), (0, 1), (1, 0)\}$$

$$\text{So } D_{R^{-1}} = \{-1, 0, 1\}$$

Q5 mathongo // mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

(I) If $(a, b) \in R_1$ and $(b, c) \in R_1$

$$\Rightarrow a^2 + b^2 \in Q \text{ and } b^2 + c^2 \in Q$$

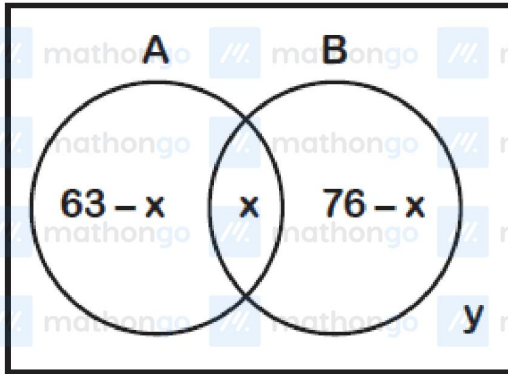
then $a^2 + 2b^2 + c^2 \in Q$ but we cannot say anything about $a^2 + c^2$, that it is rational or not.

So R_1 is not transitive

(II) If $(a, b) \in R_2$ and $(b, c) \in R_2$
 $\Rightarrow a^2 + b^2 \notin Q$ and $b^2 + c^2 \notin Q$
 but we can't say anything about $a^2 + c^2$, that it is rational or irrational.

So R_2 is not transitive

Q6



Here $63 - x + x + 76 - x + y = 100$

$$39 + y = x$$

$$\therefore 39 \leq x \leq 63$$

$\therefore$ Possible value of x is 55.

Q7

$$\bigcup_{i=1}^{50} x_i = \bigcup_{i=1}^n Y_i = T$$

$$\Rightarrow \frac{10 \times 50}{20} = \frac{5n}{6} \left\{ \begin{array}{l} \because n(x_i) = 10, n(Y_i) = 5 \\ s_0, \bigcup_{i=1}^{50} x_i = 500, \bigcup_{i=1}^n Y_i = 5n \end{array} \right\}$$

$$\Rightarrow n = 30$$

Q8

$$n(C) = 73, n(T) = 65$$

$$\therefore 65 \geq n(C \cap T) \geq 65 + 73 - 100$$

$$65 \geq x \geq 38$$

$$x \neq 36$$

Q9

The desired functions will contain either one element or two elements in its codomain of which '2' always belongs to $f(A)$.

The set B can be

$\{2\}, \{1, 2\}, \{2, 3\}, \{2, 4\}$

Total number of functions

$$= 1 + (2^3 - 2) 3$$

$$= 19$$

Q10

Number of subsets of $A = 2^m$

Number of subsets of $B = 2^n$

$$\text{Given } = 2^m - 2^n = 112$$

$$(m, n) = (7, 4)$$

$$\Rightarrow mn = 28$$